Project 2 — Bill of Materials

Document Scope

This bill of materials defines the current component structure of the manually operated adjustable workpiece fixture. It distinguishes manufactured parts, the wheel-and-handle subassembly, and the final assembly. Quantities are stated per one completed `AssemReady`.

Assembly Structure

- `AssemReady` is the final assembly and is not an individual manufactured part.
- `AssemWheel_Handle_Good` is one subassembly within `AssemReady`.
- `Wheel` and `Handle` are individual parts contained within `AssemWheel_Handle_Good`; their quantities are not additional to the subassembly quantity.
- All remaining listed parts are installed directly in `AssemReady`.

Bill of Materials

Item No.	Part / Assembly Name	Type	Quantity	Description	Main Specification	Manufacturing / Notes
A0	`AssemReady`	Final assembly	1	Complete manually operated adjustable workpiece fixture.	Final assembly definition; overall performance requirements per project requirements document.	Not manufactured as a single part. Assembly process and final inspection plan: TBD.
1	`Base_Plate`	Individual part	1	Main structural base supporting and locating the fixture components.	200 × 80 × 10 mm.	Material, tolerances, surface finish, and manufacturing method: TBD.
2	`Guide_Rod`	Individual part	2	Parallel rods that guide the linear motion of `MovingPart` and limit jaw rotation.	Ø10 mm; nominal length 150 mm each.	Material, fit, straightness, surface finish, and manufacturing method: TBD.
3	`Fixed_Jaw_New`	Individual part	1	Stationary-side jaw component used in the current assembly design.	Dimensions and interface details: TBD.	Confirm final function relative to `Fixed Jaw`; material and manufacturing method: TBD.
4	`Fixed Jaw`	Individual part	1	Additional stationary-side jaw component included in the current assembly structure.	Dimensions and interface details: TBD.	Confirm final function relative to `Fixed_Jaw_New`; material and manufacturing method: TBD.
5	`MovingPart`	Individual part	1	Translating jaw component that moves along the guide rods and engages the lead screw.	M10 × 1.5 internal threaded hole; 1.5 mm nominal travel per lead-screw revolution.	Guide-rod fits, remaining dimensions, material, tolerances, and manufacturing method: TBD.

Item No.	Part / Assembly Name	Type	Quantity	Description	Main Specification	Manufacturing / Notes
6	`NonMovingPart`	Individual part	1	Stationary support for the lead-screw mechanism.	Lead-screw support interface and remaining dimensions: TBD.	Must support screw rotation while preventing unintended axial travel; material, bearing arrangement, and manufacturing method: TBD.
7	`Lead Screw`	Individual part	1	Manually driven screw that converts rotary input into linear motion of `MovingPart`.	M10 × 1.5 external thread; nominal overall length 140 mm; pitch 1.5 mm.	Axial retention details, material, tolerances, surface finish, and manufacturing method: TBD.
8	`AssemWheel_Handle_Good`	Subassembly	1	Manual input subassembly mounted to the lead screw.	Contains one `Wheel` and one `Handle`.	Assembly connection, retention method, and inspection requirements: TBD.
8.1	`Wheel`	Individual part — child of Item 8	1	Handwheel used to apply manual torque to the lead screw.	Nominal outside diameter Ø60 mm; nominal thickness 8 mm.	Connection to `Lead Screw`, material, tolerances, finish, and manufacturing method: TBD.
8.2	`Handle`	Individual part — child of Item 8	1	Rotating grip used to turn the wheel manually.	M5 × 0.8 external thread; thread length 9 mm.	Grip geometry, material, fit, retention, finish, and manufacturing method: TBD.

BOM Control Notes

- Component names shall match the SolidWorks files exactly.
- `Lead Screw` and `MovingPart` are separate components connected through the M10 × 1.5 threaded interface.
- One full revolution of `Lead Screw` produces 1.5 mm nominal linear travel of `MovingPart`.
- Materials, finishes, tolerances, standard hardware, purchased components, and manufacturing processes shall remain `TBD` until formally selected.
- Any future fasteners, washers, bearings, retaining rings, bushings, or other hardware shall be added as separate BOM items after their specifications and quantities are defined.
- BOM quantities and component hierarchy shall be verified against the released `AssemReady` SolidWorks assembly before drawing release or manufacturing.

Current BOM Status

The component structure and listed quantities are defined for the current design state. Material selection, detailed manufacturing planning, hardware selection, tolerances, and release status remain `TBD`.